Big Basket-E-commerce-Database-Management-System

Chapter 4

4.1 Introduction:

In this phase of the **Big Basket E-Commerce Database Management System**, the relationships between attributes are studied by identifying **functional dependencies**. This helps in understanding how the data is connected and whether any information is unnecessarily repeated. The database is also analysed to identify possible **insertion, update, and deletion anomalies**. By doing this, the database can be organized efficiently, reduce data redundancy, maintain data accuracy, and prepare it for the normalization process.

4.2 Functional Dependencies:

A **functional dependency** means that one attribute (or a set of attributes) uniquely determines another attribute. In this database, the **primary key** of each entity determines all the other attributes of that entity.

Entity	**Functional Dependency**
Customer	Customer_ID →Customer Name, Email, Phone_No , DOB and Address
Category	Category_ID → Category_Name, Description
Supplier	Supplier_ID → Supplier_Name, Contact_No, Address
Product	Product_ID → Product_Name, Category_ID, Supplier_ID, Brand, Price, Stock, Expiry_Date
Cart	Cart_ID → Customer_ID, Total_Items, Total_Amount
Cart item	Cart_item_ID→ Cart_ID, Product_ID, Quantity

Order	Order_ID → Customer_ID, Order_Date, Order_Status, Total_Amount
Order item	Order_item_ID → Order_ID , Product_ID , Quantity , Price
Payment	Payment_ID → Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount
Review	Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date
Delivery	Delivery_ID → Order_ID, Delivery_Address, Delivery_Date, Delivery_Status

4.3 Redundancy Analysis:

Redundancy refers to the unnecessary duplication of data in a database. In the Big Basket E-Commerce Database, redundancy is minimized by storing each type of information in separate tables and connecting them using primary and foreign keys.

Entity	**Redundancy Analysis**
Customer	Customer details are stored only once and referenced using Customer_ID.
Category	Category details are stored separately using Category_ID.
Supplier	Supplier information is stored once and linked using Supplier_ID.
Product	Product details are stored once and related tables use foreign keys.
Cart	Cart information is maintained separately for each customer.

Cart_Item	Only Product_ID and Quantity are stored, avoiding duplicate product details.
Order	Order details are stored separately and linked to Customer_ID.
Order_Item	Product information is referenced using Product_ID instead of duplication.
Payment	Payment details are stored separately for each order.
Review	Reviews store only references to Customer and Product.
Delivery	Delivery information is stored separately and linked to Order_ID.

4.4 Data Anomalies:

Data anomalies are problems that occur when data is duplicated or not properly organized in a database. The three common types of anomalies are insertion anomaly, update anomaly, and deletion anomaly. The Big Basket E-Commerce Database is designed using separate tables and primary and foreign keys, which helps minimize these anomalies.

Entity	Insert Anomaly	Update Anomaly	Delete Anomaly
Customer	A new customer can be added without creating an order	Customer details(eg. Phone_no) are updated only in the customer table	Deleting an order does not delete customer information.
Category	A new category can be added without	Category name is updated only in the	Deleting a product does not remove the

	adding a product.	Category table.	category.
Supplier	A new supplier can be added without adding a product.	Supplier contact details are updated only in the Supplier table.	Deleting a product does not remove supplier information.
Product	A new product can be added before any order is placed.	Product price or stock is updated only in the Product table.	Deleting an order does not delete the product.
Cart	A cart can be created for a customer independently.	Cart information is updated only in the Cart table.	Deleting a cart does not delete customer information.
Cart_Item	Products can be added to a cart independently.	Quantity is updated only for the selected cart item.	Removing a cart item does not delete the product.
Order	An order can be created for an existing customer.	Order status is updated only in the Order table.	Deleting an order does not delete customer or product details.
Order_Item	Products can be added to an order independently.	Quantity or price changes affect only that order item.	Deleting an order item does not remove the product.
Payment	Payment details can be added after an order is created.	Payment status is updated only in the Payment table.	Deleting a payment record does not remove the order.
Review	A customer can submit a review after purchasing a product.	Rating or review text is updated only in the Review table.	Deleting a review does not delete the product or customer.
Delivery	Delivery details can be added after an order is confirmed.	Delivery status is updated only in the Delivery table.	the Delivery table. Deleting delivery information does not remove the

			order.